

Douglas Gordon Daly

Principal AI Architect / AI Platform Engineering Lead

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PROFESSIONAL SUMMARY

Senior technical leader with deep expertise architecting enterprise AI/ML platforms, governed GenAI and tool-calling systems, and operational intelligence solutions. Built governed analytics assistants, monitoring frameworks, and reusable data/AI platforms with strong emphasis on controls, observability, and auditable execution. Experienced across cloud-native AI infrastructure, large-scale data platforms, executive alignment, and technical mentorship. Known for translating ambiguous business needs into scalable, reliable analytical systems that improve decision-making and operational performance.

CORE EXPERTISE

- AI Platform Architecture - enterprise AI platforms, governed analytics assistants, operational intelligence systems, reusable AI/analytics frameworks
- GenAI & Tool-Calling Systems - LLM applications, structured outputs, tool orchestration, RAG/GraphRAG concepts, prompt engineering, evaluation, traceability
- Governance & Operational Controls - responsible AI adoption, guardrails, auditable execution, analyst-in-the-loop workflows, policy-aware system design
- Observability & Reliability - monitoring frameworks, incident detection, alerting, operational health, SLIs/SLOs, SEV analysis, on-call-friendly design
- Cloud & Data Platforms - Python, SQL, BigQuery, AWS SageMaker/S3/Terraform, GCP, Docker, data profiling, validation, cross-system integration
- Technical Leadership - architecture direction, executive alignment, technical roadmaps, stakeholder translation, mentoring, developer enablement

PROFESSIONAL EXPERIENCE

Principal Consultant | Daly Engineers LLC | Jan 2026 - Present

Consolidated Edison (via ABC Consulting)

- Architected and delivered a governed analytics assistant leveraging Gemini, BigQuery, and tool-calling architectures, embedding analyst rules and operational controls to ensure predictable, auditable, and compliant execution of generative AI workflows.
- Developed automated frameworks for profiling, type inference, duplicate detection, candidate key analysis, and cross-system join validation across 100+ source tables and millions of operational and financial records, enhancing data quality and interoperability.
- Defined a phased analytical maturity roadmap adopted as the foundation for anomaly detection, construction analytics, and AI-enabled decision support initiatives.
- Led analytical readiness assessment for Oracle/EBS and Maximo data migration to BigQuery, establishing the groundwork for cross-domain analytics spanning construction operations and financial systems.

NBCUniversal (via Apex Systems)

- Architected AWS-based AI and graph analytics pipelines leveraging SageMaker, Terraform, S3, and containerized workloads; evaluated graph analytics and graph machine learning solutions including Neptune, GraphStorm, Bedrock, and LangGraph to support dependency analysis and operational intelligence initiatives.

AI Instructor | SimpliLearn | October 2025 - Present

- Mentored students on AI architecture, governance, implementation tradeoffs, and enterprise adoption strategies, enabling cross-functional technical fluency and responsible AI practices.
- Delivered instruction on transformer architectures, LLMs, RAG, agentic and multimodal AI, diffusion models, and enterprise AI governance, fostering advanced understanding of generative and agentic AI technologies.

AI Instructor | Interview Kickstart | March 2025 - Present

- Mentored students on AI architecture, implementation patterns, and technology selection tradeoffs, fostering technical fluency and cross-team knowledge sharing.
- Delivered instruction on practical applications of LLMs, retrieval-augmented generation (RAG), agentic AI, and modern AI development workflows to enable next-generation AI solution development.

Data Scientist | Intuit (via The Cydio Group) | Apr 2025 - Jul 2025

- Partnered with finance, marketing, and product stakeholders to translate attribution and performance data into actionable business recommendations, driving cross-functional alignment between business needs and technical solutions.
- Consolidated fragmented reporting into unified Tableau dashboards and automated executive reporting workflows using Python and SQL, reducing report preparation time from days to hours and enhancing data integration and process efficiency.

Data Scientist | Meta | October 2021 - January 2025

- Designed and developed the ReMon framework, empowering engineering teams to implement robust monitoring and alerting systems without deep internal tooling expertise; adopted across Meta FinTech and instrumental in the automatic detection of over 15 SEVs in 2024.
- Developed forecasting and performance-management frameworks that translated engineering initiatives into projected reliability outcomes, supporting planning, goal setting, and leadership decision-making across multiple business pillars.

Data Scientist | Samsung Electronics (via Harvey Nash) | Dec 2020 - Oct 2021

- Developed and deployed gradient-boosting propensity models and automated audience segmentation workflows, driving a 146% increase in campaign conversions and a 229% rise in campaign-attributed revenue while improving efficiency and scalability of digital marketing initiatives.

Head of Data Science | Facticeus | Oct 2017 - Nov 2020

- Led data science initiatives on a 30B+ transaction dataset, architecting analytical products and market-intelligence capabilities through large-scale feature engineering, merchant intelligence, and consumer-spending analytics.

Expert SEO Solutions Architect | Nike (via Intersoft Inc.) | Nov 2016 - Sep 2017

- Architected a scalable SEO and information-architecture framework for Nike's product discovery, aligning customer navigation with search engine indexation to support a rapidly growing product catalog.

Senior Manager, Operations Analysis | Capital One | Jan 2012 - Dec 2016

- Conceived, architected, and led adoption of RITHM, an operational intelligence platform aggregating real-time monitoring signals by business capability and system ownership, reducing major incident detection times from hours to minutes.
- Secured executive sponsorship and partnered with CTO-level leadership to define enterprise monitoring strategy, operational health metrics, and reliability initiatives, expanding Splunk-based prototypes into a scalable Kafka and Elastic Stack solution aligned with open-source strategy.

Senior Principal Systems Engineer / Program Manager | Raytheon | Mar 2005 - Dec 2011

- Led systems engineering organizations of 40-50 engineers specializing in modeling, simulation, analysis, and advanced aerospace systems, providing technical leadership, mentorship, and proposal support for programs valued up to \$50M.
- Turned around an over-budget classified fixed-price technology program by establishing execution plans, aligning engineering teams, and proactively managing technical and program risks; delivered under budget and nearly doubled the original profit target on a historically challenging delivery effort.

SELECTED AI PROJECTS

GraphRAG Engine & LoRA Coverage Assistant | GitHub | 2025

- Built a GraphRAG system for multi-hop benefits reasoning with fully traceable retrieval paths across entities, clauses, and supporting evidence.
- Fine-tuned a LoRA adapter on a 7B open-source LLM to enforce a strict JSON schema (decision/reason/disclaimer) for reliable downstream automation.

Agent-to-Agent Protocol MVPs | GitHub | 2026

- Agent Olympics: built a small evaluation runner that benchmarks a generalist agent vs discovered specialists across three task categories with explicit scoring criteria and summary reporting.
- AI PEMDAS: built an instrumented coordination workflow where an evaluator routes sub-steps to ADD vs MULT agents and logs calls for end-to-end traceability (FastAPI JSON-RPC).

Agentic Multimodal Orchestrator | GitHub | 2025

- Orchestrated structured retrieval, layout planning, and diffusion prompting to generate data-driven posters/maps with repeatable, template-based workflows.
- Implemented template constraints and parameterized prompts to ensure consistency across runs.

EDUCATION

MBA | (UCLA Anderson School of Management): Top Honors | GPA: 3.9

M.S., Applied Statistics | (Rochester Institute of Technology): GPA: 3.9

M.S., Electrical Engineering | (University of Washington): GPA: 3.7

B.S., Physics (Minor: Mathematics) | (Humboldt State University): GPA: 3.4

PATENTS & RECOGNITION

- Lead Inventor, U.S. Patent #8635622: Method and System for Resource Management Using Fuzzy Logic Timeline Filling (2014)
- Capital One Business Technology Innovation Award (2014)
- Conference Chair & Speaker - Customer Analytics Innovation Summit (2016)